Exact-partition failure at the egalitarian core: a complete boundary phase diagram

Maxwell Flanagan

Draft of August 4, 2026

Abstract

For a transferable-utility game, exact partition at an allocation means that every cumulative union of its richest payoff strata is tight. This condition and its Lorenz consequence are established. We determine exactly when the condition can fail at the leximin core allocation of a singleton-normalized exact game. If a strict payoff boundary has m cumulative poorer players and h richer players, then a slack richer suffix exists if and only if

$$m \geq m_{\min}(h), \qquad m_{\min}(1) = 4, \quad m_{\min}(2) = m_{\min}(3) = 3, \quad m_{\min}(h) = 2 \ (h \geq 4).$$

The lower bounds are structural. Leximin optimality gives a strict cover by tight coalitions; exactifying the failed suffix turns that cover into a transport problem through minimal balanced hypergraphs. One poorer player cannot support the required opposing signs. A unique richest player needs four poorer players by a fractional sacrifice budget. With two poorer players, every failure needs a proper active hyperedge of size at least three, hence at least four richer players. Three explicit dimension-free lower-envelope families attain every remaining pair. The familiar four-player guarantee and sharp five-player failure are immediate corollaries of this two-parameter theorem. No finite enumeration is used in the proof.

1 Introduction and prior-art boundary

Let (N, v) be a transferable-utility game and

$$\mathrm{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for every } S \subseteq N\}.$$

The game is *exact* if every coalition inequality is attained by some core allocation. Write $\phi = \mathrm{Lmin}(v)$ for the unique core allocation whose increasing rearrangement is lexicographically maximal. At a strict payoff boundary of ϕ, let L be the cumulative poorer side and let $H = N \setminus L$ be the richer suffix.

The exact-partition condition and its implications for Dutta–Ray and Lorenz selection are prior art (Llerena and Mauri, 2017; Sánchez-Soriano et al., 2010). Property I and its antidual form provide established qualitative tight-coalition descriptions of the egalitarian selections (Arin et al., 2008; Mauri, 2018). Exact balancedness is also established (Csóka et al., 2011; Lohmann et al., 2012). Our candidate contribution is the sharp two-parameter classification, its transport proof, and the realizing families. The closest published low-dimensional result known to us is that exact games through four players are PES stable (Dietzenbacher, 2021); that does not itself assert exact partition at Lmin.

The main theorem is stated now. All games below are singleton-normalized, meaning $v(\{i\}) = 0$ for every player.

Theorem 1 (boundary phase diagram)**.** *At a strict payoff boundary of $\phi = \mathrm{Lmin}(v)$ in a singleton-normalized exact game, put $m = |L|$ and $h = |H|$. It is possible that $v(H) < \phi(H)$ exactly for*

$$m \geq m_{\min}(h), \qquad m_{\min}(h) = \begin{cases} 4, & h = 1, \\ 3, & h = 2 \text{ or } 3, \\ 2, & h \geq 4. \end{cases} \tag{1}$$

Every admitted pair is realized by an integral indecomposable exact game. The poorer side may contain several payoff strata.

The proof separates a general lower-bound mechanism from three universal constructions. This makes the four/five threshold a corollary rather than the theorem's organizing principle.

2 Strict covers and balanced-trace transport

For $\phi \in \mathrm{Core}(v)$ write

$$s_\phi(S) = \phi(S) - v(S).$$

A coalition is ϕ-tight when its slack is zero.

Lemma 2 (strict richer cover)**.** *Let $\phi = \mathrm{Lmin}(v)$ and let $N = L \dot\cup H$ be a strict payoff boundary. There are nonnegative weights on proper ϕ-tight coalitions whose incidence load is a common positive number on H and is strictly smaller at every player in L.*

Proof. This is the Farkas-dual form of the established transfer-blocking/property-I description of the egalitarian core. For completeness, if no such cover exists, separation gives an efficient tangent direction that raises the first affected richer-side cumulative level without lowering an earlier payoff stratum. A sufficiently short step from ϕ remains in the core and lexicographically improves its increasing rearrangement, contradicting the definition of Lmin. □

Normalize the common rich load in Lemma 2 to one and remove coalitions disjoint from H. Project every remaining tight coalition to its nonempty trace on H. The projected weights lie in the exact-cover polytope

$$\mathcal{P}_H = \left\{ \beta \geq 0 : \sum_{\emptyset \neq R \subseteq H} \beta_R \mathbf{1}_R = \mathbf{1}_H \right\}. \tag{2}$$

Its vertices are the unique normalized weights of minimal balanced collections on H; this is standard balanced-collection geometry (Lohmann et al., 2012; Laplace Mermoud et al., 2025).

Theorem 3 (balanced-trace transport)**.** *Suppose $s = s_\phi(H) > 0$. There are a core point z, a displacement $\Delta = z - \phi$, and a finite convex mixture of lifted vertices of $\mathcal{P}_H$ with poor-load vectors $r^q \in \mathbb{R}^L_+$ such that*

$$\Delta(H) = -s, \qquad \Delta(L) = s, \tag{3a}$$

$$\min_{i \in L} \Delta_i < 0 < \max_{i \in L} \Delta_i, \tag{3b}$$

$$-s + \sum_{i \in L} r^q_i \Delta_i \geq 0 \qquad \text{for every } q, \tag{3c}$$

$$\sum_q \theta_q r^q_i < 1 \qquad \text{for every } i \in L. \tag{3d}$$

Proof. Exactness supplies $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$, proving (3a). Let $c_i < 1$ be the poorer loads of a normalized cover from Lemma 2. Every supported coalition is tight at ϕ and feasible at z, hence

$$0 \le \sum_T \lambda_T \Delta(T) = -s + \sum_{i \in L} c_i \Delta_i. \tag{4}$$

If every poorer displacement were nonnegative, the right side would be strictly smaller than $-s + \Delta(L) = 0$. Since $\Delta(L) = s > 0$, both signs occur, proving (3b).

Decompose the projected cover in (2) into vertices. For each trace, split its weight among the original tight coalitions having that trace; taking the product of these conditional choices gives a finite mixture of lifted packets. Summing the tight-coalition inequalities at z inside one packet gives (3c), while averaging the lifted packets recovers the original poorer loads $c_i < 1$, proving (3d). □

The first exclusion is immediate.

Corollary 4. *A slack boundary cannot have $m = 1$.*

Proof. Equation (3b) requires both a positive and a negative displacement on L. □

3 The two remaining lower bounds

3.1 A unique richest player

Assume $H = \{r\}$, write $\phi_r = \gamma$, and put $q_i = \phi_i < \gamma$ for $i \in L$. The transfer-blocking characterization gives, for each $i \in L$, a tight protector

$$T_i = \{r\} \cup Q_i, \qquad \emptyset \ne Q_i \subseteq L \setminus \{i\}. \tag{5}$$

The nonemptiness follows because the normalized singleton $\{r\}$ is slack.

Lemma 5 (sacrifice budget)**.** *For weights $\lambda_i \ge 0$ satisfying*

$$c_j := \sum_{i : j \in Q_i} \lambda_i \le 1, \qquad \Lambda := \sum_i \lambda_i,$$

one has

$$\gamma(\Lambda - 1) \le \sum_{j \in L} (1 - c_j) q_j. \tag{6}$$

Proof. Exactify $\{r\}$ at a core point z and put $d = z - \phi$. Summing the tight protector inequalities with the displayed weights, using efficiency and the singleton lower bounds $z_j \ge 0$, gives (6). Equivalently, complete the uncovered poorer loads by singleton coalitions; together with negative weight $\Lambda - 1$ on $\{r\}$, the incidence vector is exact-balanced, and the exact-balanced slack inequality is exactly (6). □

Proposition 6. *If $h = 1$ and the suffix is slack, then $m \ge 4$.*

Proof. For $m = 1$ no protector in (5) exists. For $m = 2$, both protector traces are the opposite singleton; weight both by one in (6), obtaining $\gamma \le 0$.

Let $m = 3$. If some $Q_i = \{j\}$, weight the protectors indexed by i and j by one. Their traces are disjoint and leave at most one player u uncovered, so (6) gives either $\gamma \le 0$ or $\gamma \le q_u < \gamma$. If no Q_i is a singleton, then $Q_i = L \setminus \{i\}$ for all i. Weight all three protectors by $1/2$; every poorer player has load one and (6) gives $\gamma/2 \le 0$. □

3.2 Exactly two poorer players

We first isolate the exact-face inequality used in the rounding argument.

Lemma 7 (exact-face overload)**.** *Let v be singleton-normalized and exact, let $\phi \in \mathrm{Core}(v)$, and let $D \subseteq H \subsetneq N$. Suppose nonnegative weights on proper ϕ-tight coalitions have load $1+q$ on D, load one on $H \setminus D$, and loads $0 \le c_i \le 1$ on $i \in N \setminus H$, where $q > 0$. Then*

$$qs_\phi(D) \le \sum_{i \notin H} (1 - c_i)\phi_i. \tag{7}$$

Proof. Choose $x \in \mathrm{Core}(v)$ with $x(D) = v(D)$ and put $d = x - \phi$. Summing the supported tight-coalition inequalities at x gives

$$0 \le d(H) + qd(D) + \sum_{i \notin H} c_i d_i.$$

Efficiency replaces $d(H)$ by $-\sum_{i \notin H} d_i$. Hence

$$qs_\phi(D) = -qd(D) \le \sum_{i \notin H} (1 - c_i)(-d_i) \le \sum_{i \notin H} (1 - c_i)\phi_i,$$

where the last inequality uses $x_i \ge v(\{i\}) = 0$. □

Put $L = \{a, b\}$ and label the displacement in Theorem 3 so that $\Delta_a < 0 < \Delta_b$. By (3d), some lifted extreme packet has $r_b < 1$.

Lemma 8 (proper-hyperedge obstruction)**.** *Every such underloaded packet contains a proper rich trace R with*

$$3 \le |R| \le |H| - 1. \tag{8}$$

Proof. Suppose instead that every proper supported trace has size at most two. The positive incidence columns of a vertex of (2) are independent. Its rank-two components are therefore isolated singletons of weight one, isolated edges of weight one, and vertex-disjoint odd cycles whose edges have weight $1/2$. This follows by the usual leaf deletion: a positive singleton cannot share a nontrivial component; a tree reduces to one edge; an even cycle is dependent; and an odd cycle has the unique half cover.

An integral packet has integer b-load, and (3c) forces that load to be at least one. Hence the selected packet is fractional. Its poor loads are half-integers. From

$$-s + r_a\Delta_a + r_b\Delta_b \ge 0, \qquad \Delta_a + \Delta_b = s,$$

and $r_b < 1$, one obtains $r_b = 1/2$ and $r_a = 0$. Exactly one half-weight edge on one odd cycle therefore carries b, and no supported coalition carries a.

Round that labeled odd cycle alternately, starting with the labeled edge, and leave every other component at its original weight. The resulting nonnegative combination of proper ϕ-tight coalitions has rich load $\mathbf{1}_H + e_d$ for one rich player d, poor load e_b, and no load on a. Exactify singleton d. Summing the selected tight inequalities and adding the omitted singleton a gives the exact-face overload inequality $\phi_d \le \phi_a$, contradicting the strict boundary. Indeed, Lemma 7 has $D = \{d\}$, $c_b = 1$, and $c_a = 0$. Thus (8) holds. □

Proposition 9. *If $m = 2$ and the suffix is slack, then $h \ge 4$.*

Proof. Lemma 8 requires a proper trace of size at least three. □

The proof so far yields every forbidden pair in (1) without enumerating games or payoff-stratum patterns.

4 Three universal realizing families

We use a standard lower-envelope observation.

Lemma 10 (lower-envelope realization)**.** *Let G be a finite family of nonnegative allocations with common total T, and define $v(S) = \min_{g\in G} g(S)$. Then every $g \in G$ belongs to* Core(v)*, every coalition attains its worth, and v is exact. If every coordinate is zero in some row, then v is singleton-normalized.*

Proof. The defining inequality gives $g(S) \geq v(S)$ for every g, S, so every row is a core allocation. Conversely every coalition minimum is attained at a row. The last assertion is immediate. □

We record the families compactly. The displayed face inequalities give short progressive leximin proofs; no search is involved.

Proposition 11 (unique-richest family)**.** *For every $m \geq 4$ there is an integral indecomposable exact game with m poorer players $P = \{1, \dots, m\}$, one richer player r, and*

$$\mathrm{Lmin}(v) = \mathbf{1}_P + (m-2)e_r, \qquad v(\{r\}) = 0.$$

Proof. Apply Lemma 10 to the four common-total rows

$$\begin{aligned} g^1 &= (m-1)(e_4 + e_r),\\ g^2 &= (m-1)(e_3 + e_4),\\ g^3 &= e_1 + e_2 + e_3 + (m-4)e_5 + (m-1)e_r,\\ g^4 &= \mathbf{1}_P + (m-2)e_r, \end{aligned}$$

where the zero e_5 term is omitted at $m = 4$. Every coordinate vanishes in a row. The four attained faces

$$v(N \setminus \{1\}) = v(N \setminus \{2\}) = 2m - 3, \qquad v(\{3, r\}) = v(\{4, r\}) = m - 1$$

show that any core point giving everyone at least one fixes players $1, 2$ at one; subtracting $\mathbf{1}_N$ and comparing the two pair inequalities with efficiency then fixes every poorer player at one. The remaining total gives $x_r = m - 2$. Hence $g^4 = \mathrm{Lmin}(v)$. Row g^2 proves $v(\{r\}) = 0$. For indecomposability, observe that

$$v(\{1, 4\}) = v(\{2, 4\}) = v(\{3, 4\}) = 1, \qquad v(\{3, r\}) = v(\{4, r\}) = m - 1,$$

and, when $m \geq 5$, $v(\{4, 5\}) > 0$. These positive pair worths place players $1, \dots, 5, r$ in one direct-sum component. If $m = 6$, the coalition $T = \{3, 4, 5, r\}$ satisfies $v(T) = 7 < v(T \cup \{6\}) = 8$. If $m \geq 7$, put $T = \{4, 5\}$; then $v(T) = 2 < v(T \cup \{j\}) = 3$ for every $j \geq 6$. Since every remaining singleton has value zero, these strict interactions attach every remaining player to the same component. □

Proposition 12 (two-poorer internal-sacrifice family)**.** *For every $m \geq 2$ and $h \geq 4$ there is an integral indecomposable exact game with a poorer block P of size m, a richer suffix H of size h, and*

$$\mathrm{Lmin}(v) = D\mathbf{1}_P + 2D\mathbf{1}_H, \qquad v(H) = 2D(h-1), \qquad D = (h-2)(h-3).$$

Proof. Put $r = h - 2$, write $P = \{p_1, p_2\} \dot\cup Q$ and $H = \{a, b\} \dot\cup K$, $|K| = r$. Use the common-total rows

$$\begin{aligned}
u &= De_{p_2} + D\mathbf{1}_Q + 3D(e_a + e_b) + (2r-1)(r-1)\mathbf{1}_K,\\
z &= 4De_{p_1} + D\mathbf{1}_Q + 2(r^2-1)\mathbf{1}_K,\\
w^k &= De_{p_1} + D\mathbf{1}_Q + 3D(e_a + e_b) + r(2r-1)\mathbf{1}_{K\setminus\{k\}} \quad (k \in K),\\
\phi &= D\mathbf{1}_P + 2D\mathbf{1}_H,\\
q^j &= \phi + D(e_{p_1} - e_j) \quad (j \in Q).
\end{aligned}$$

Lemma 10 gives a normalized integral exact game. Direct substitution yields

$$\begin{aligned}
v(N \setminus \{p_2\}) &= D(m + 2h) - D,\\
v(H \setminus \{a\}) = v(H \setminus \{b\}) &= 2D(h-1),\\
v(\{p_1, a, b\} \cup (K \setminus \{k\})) &= D(2h-1),\\
v(H) &= 2D(h-1).
\end{aligned}$$

At a core point x above the floor D, the complement face first fixes $x_{p_2} = D$. Write $y = x - D\mathbf{1}_N$, $E = y(Q)$, and $t = y(K)$. Then $y \geq 0$, $y_{p_2} = 0$, and $y(N) = hD$. The two co-singleton faces give

$$y_a + t \geq (h-1)D, \qquad y_b + t \geq (h-1)D. \tag{9}$$

The $r = h - 2$ remaining faces give, for every $k \in K$,

$$y_{p_1} + y_a + y_b + t - y_k \geq (h-1)D.$$

Summing them and substituting $y_{p_1} + E + y_a + y_b + t = hD$ yields $t + rE \leq rD$. On the other hand, (9) and efficiency imply

$$y_{p_1} + E \leq t - rD \leq -rE.$$

Nonnegativity forces $y_{p_1} = E = 0$. Hence all of P is pinned at level D; efficiency leaves total $2Dh$ on H, equalized by ϕ. Thus $\phi = \mathrm{Lmin}(v)$ and its richer suffix has slack $2D$. The same construction is indecomposable. Indeed, p_1 has positive pair worth with every other player:

$$v(\{p_1, p_2\}) = v(\{p_1, j\}) = v(\{p_1, k\}) = D \quad (j \in Q, k \in K), \qquad v(\{p_1, a\}) = v(\{p_1, b\}) = 3D.$$

Thus the positive-pair graph is connected. □

Proposition 13 (three-poorer escape family)**.** *For every $m \geq 3$ and $h \geq 2$ there is an integral indecomposable exact game with m cumulative poorer players and richer suffix H, $|H| = h$, such that*

$$\mathrm{Lmin}(v) = \mathbf{1}_{\{p_1,p_2\}\cup Q} + 2e_{p_3} + 3\mathbf{1}_H, \qquad v(H) = 3h - 1, \qquad |Q| = m - 3.$$

Proof. Write $H = \{r_1, \ldots, r_h\}$ and take the lower envelope of

$$\begin{aligned}
a &= 2e_{p_1} + 3e_{p_3} + \mathbf{1}_Q + 5e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1,r_h\}},\\
b &= 2e_{p_2} + e_{p_3} + \mathbf{1}_Q + 4e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1\}},\\
c &= \mathbf{1}_Q + (3h+4)e_{r_h},\\
\phi &= e_{p_1} + e_{p_2} + 2e_{p_3} + \mathbf{1}_Q + 3\mathbf{1}_H,\\
q^j &= \phi + e_{p_1} - e_j \quad (j \in Q).
\end{aligned}$$

All rows have total $m+3h+1$, and every coordinate vanishes in a row, so Lemma 10 applies. The attained values

$$\begin{aligned} v(N \setminus \{j\}) &= m+3h \quad (j \in Q), \\ v(H \cup \{p_3\}) &= 3h+2, \\ v(H \cup \{p_1\}) &= 3h+1, \\ v(H) &= 3h-1 \end{aligned}$$

give the progressive proof. On the face where every player receives at least one, the complement faces fix Q, the second face fixes p_1, p_2 at one, and the third fixes p_3 at two. Efficiency leaves total $3h$ on H, equalized at three by ϕ. Thus $\phi = \mathrm{Lmin}(v)$ and $s_\phi(H) = 1$. When $Q \neq \emptyset$, one member of Q has positive pair worth with every other player. When $Q = \emptyset$, the positive-pair graph connects all but p_2, and

$$v(\{p_1, p_3, r_h\}) = 4 < 5 = v(\{p_1, p_2, p_3, r_h\})$$

attaches p_2. Hence the game is indecomposable. □

Proof of Theorem 1. Corollary 4 excludes $m = 1$ for every h. Proposition 6 excludes $m \leq 3$ when $h = 1$, and Proposition 11 realizes every $m \geq 4$. Proposition 9 excludes $h \leq 3$ when $m = 2$, and Proposition 12 realizes every $m \geq 2, h \geq 4$. All remaining pairs have $m \geq 3, h \geq 2$ and are realized by Proposition 13. □

5 The four/five threshold and its mechanism

Corollary 14 (sharp player threshold)**.** *Every singleton-normalized exact game on at most four players is exact partition at* Lmin*. Five players suffice for failure.*

Proof. Every pair (m, h) with $m + h \leq 4$ is forbidden by (1). The pair $(m, h) = (3, 2)$ is realized by Proposition 13 on five players. Alternatively, $(4, 1)$ is realized by Proposition 11. □

The theorem gives two distinct first mechanisms. The $(4, 1)$ obstruction is an exterior sacrifice: the fourth poorer player supplies the first positive singleton budget capable of financing a slack richest singleton. The $(3, 2)$ obstruction is a mixed block-star mechanism. Two-poorer failures do not yet exist on five players: their active balanced packet must contain a proper rank-three trace, which first fits at $(2, 4)$ on six players. Thus "five" is not an unexplained output of enumeration; it is the first total size at which either of the two admissible boundary regions in (1) is entered.

6 What the theorem does and does not establish

The phase theorem is dimension-free in the sense relevant to the original problem: it classifies every possible pair of boundary-side cardinalities, and its lower bounds are symbolic transport arguments. It does not classify all support-minimal balanced packets or all extreme rays of the associated two-sided coupling cone. Distinct irreducible packet classes already occur: the co-singleton class $J - I$ and a blown-up three-cycle class are not related by a single universal minor. A full obstruction theory would therefore need a multi-class circuit or coupling characterization rather than a claim that every failure contains one fixed small configuration.

The theorem's novelty remains conditional on specialist review. Exact partition, its Lorenz consequence, property I, exact balancedness, minimal balanced collections, and lower-envelope exactification are not claimed. The candidate new content is the Lmin-selected transport bridge, the sharp staircase (1), and the three dimension-free indecomposable realizing families.

Reproducibility note

The proof above is human and does not depend on finite enumeration. Exact rational programs in the companion repository independently reconstruct the smallest family members, all coalition values, and staged Lmin solutions. They are bug audits, not proof evidence.

References

Arin J, Iñarra E (2001) Egalitarian solutions in the core. International Journal of Game Theory 30:187–193.

Arin J, Kuipers J, Vermeulen D (2008) An axiomatic approach to egalitarianism in TU-games. International Journal of Game Theory 37:565–580. https://doi.org/10.1007/s00182-008-0133-6

Csóka P, Herings PJJ, Kóczy LÁ (2011) Balancedness conditions for exact games. Mathematical Methods of Operations Research 74:41–52.

Dietzenbacher B (2021) Procedural values for cooperative games. Doctoral thesis, Tilburg University.

Llerena F, Mauri L (2017) On the existence of the Dutta–Ray's egalitarian solution. Mathematical Social Sciences 89:92–99. https://doi.org/10.1016/j.mathsocsci.2017.07.002

Laplace Mermoud L, Grabisch M, Sudhölter P (2025) Minimal balanced collections and their applications to core stability and other topics of game theory. Working paper.

Lohmann E, Borm P, Herings PJJ (2012) Minimal exact balancedness. Mathematical Social Sciences 64:127–135.

Mauri Masdeu L (2018) Essays on egalitarianism-based solution concepts for cooperative TU-games. Doctoral thesis, Universitat Rovira i Virgili.

Sánchez-Soriano J, Brânzei R, Llorca N, Tijs SH (2010) A technical note on Lorenz dominance in cooperative games. CentER Discussion Paper 2010–101. https://pure.uvt.nl/ws/files/1269376/2010-101.pdf